

RPG-Massey Hypothetic Research Plan for formal validation

The two texts are deeply and positively connected. The main essay (Text 1) provides the broad structural landscape of Real Projective Geometry, culminating in Section 6.3 with the topological properties of $\mathbb{RP}^n$, specifically its embedding and immersion bounds. It briefly states that $\mathbb{RP}^n$ immerses in $\mathbb{R}^{2n-\alpha(n)}$, citing Cohen (1985).

Text 2 acts as a deep-dive “zoom-in” on this exact statement. It explains *why* this specific dimension $2n - \alpha(n)$ is the magic number: the Stiefel-Whitney classes of the normal bundle vanish above this dimension for *all* manifolds (Massey's obstruction), and **products of real projective spaces are the extremal examples that prove this bound is sharp**. In other words, real projective geometry is not just an application of Massey's conjecture; it is the fundamental test case that defined the conjecture's lower bound.

To generate a formal validation of their connected content using **Lean 4**, one must recognize the current state of Lean's mathematical library (mathlib4). While basic differential geometry (manifolds, tangent bundles) is well-developed, advanced algebraic topology (Stiefel-Whitney classes, Brown-Peterson homology, surgery theory) is largely uncharted territory.

Therefore, an initial plan for formal validation must be phased, starting from the foundations and building up to the formulation (and eventually, the proof) of the conjecture. Here is a strategic blueprint for formalizing this content in Lean 4.

Phase 1: Foundational Geometry and Arithmetic (Highly Feasible, Short-Term)

Before doing topology, we must formalize the basic objects.

1. **The Function $\alpha(n)$:** Formalize the function that counts the number of 1s in the binary representation of n . This is purely computational and easy in Lean 4.

```
def alpha (n : ℕ) : ℕ := (Nat.digitBases 2 n).sum -- conceptual pseudo-Lean
-- Validate basic properties: alpha(2^k) = 1, etc.
```

2. **Definition of $\mathbb{RP}^n$:** Lean's mathlib4 already has the Projectivization of a vector space. We need to instantiate it for $\mathbb{R}^{n+1}$ and formally endow it with a SmoothManifoldWithCorners structure using the standard atlas mentioned in Text 1 (Section 1.1).
3. **Immersions:** Formalize the concept of an immersion (a smooth map with an injective differential at every point). This relies on Lean's ContMDiff and HasMFDeriv structures.

Phase 2: Vector Bundles and Cohomology (Medium-Term, Mathlib PR level)

To connect Text 1's Stiefel-Whitney calculations to Text 2's obstruction theory, we need characteristic classes.

1. **The Tangent Bundle of $\mathbb{RP}^n$:** Formalize the tautological line bundle γ^1 and formally prove the Euler sequence: $T\mathbb{RP}^n \oplus \mathbf{1} \cong (n+1)(\gamma^1)^*$.
2. **Stiefel-Whitney Classes:** Define the Stiefel-Whitney classes $w_i(E) \in H^i(X; \mathbb{Z}_2)$ for a real vector bundle E . This requires setting up singular cohomology with $\mathbb{Z}_2$ coefficients and proving the Whitney sum formula: $w(E \oplus F) = w(E) \smile w(F)$.
3. **The Total Class of $\mathbb{RP}^n$:** Formally verify the equation from Text 1, Section 6.2: $w(\mathbb{RP}^n) = (1 + a)^{n+1}$.

Phase 3: Formulating the Statements (Medium-Term)

Even without proving Cohen's theorem, *stating* Massey's theorem and conjecture rigorously in Lean 4 is a highly valuable exercise in formalization.

1. Stating Massey's Theorem (The Obstruction):

```
theorem massey_obstruction {n : ℕ} (M : SmoothManifold ℝ n) (hM : CompactSpace
  ↪ M) :
  ∀ i > n - alpha n, w i (stable_normal_bundle M) = 0
```

2. Proving the Sharpness (The Projective Connection):

Formalize the proof that the bound cannot be lowered by calculating the classes for $\mathbb{RP}^{2^k}$. Since $w(\mathbb{RP}^{2^k}) = (1 + a)^{2^k+1} = 1 + a + a^{2^k}$, the class w_{2^k} is non-zero, creating a formal obstruction to immersing $\mathbb{RP}^{2^k}$ in $\mathbb{R}^{2^{k+1}-2}$.

3. Stating Cohen's Theorem (The Resolution):

```
theorem cohen_immersion (n : ℕ) (M : SmoothManifold ℝ n) (hM : CompactSpace M)
  ↪ :
  ∃ (d : ℕ) (f : M → EuclideanSpace ℝ d), d = 2 * n - alpha n ∧ IsImmersion f
```

Phase 4: Blueprinting the Proof (Long-Term, Research Level)

Proving Cohen (1985) in Lean 4 would be a massive undertaking, comparable to or exceeding the celebrated *Sphere Eversion Project* (which formalized Smale's paradox in Lean 3). It would require the creation of a **Lean Blueprint** (using the `leanblueprint` tool developed by Patrick Massot), which turns a LaTeX document into a dependency graph for formalization.

The blueprint graph would need the following major nodes:

1. **Cobordism Theory:** Defining the unoriented cobordism ring $\mathfrak{N}_*$ (Thom's theorem).
2. **Surgery Theory:** Formalizing the ability to perform surgery on a manifold to simplify its normal bundle while remaining in the same cobordism class.
3. **Classifying Spaces:** Constructing BO and its finite-dimensional approximations $BO(k)$.
4. **Brown-Peterson Homology:** Defining the spectrum BP and formalizing the BP-homology of BO .

Summary of the Formalization Pipeline

If you were to launch this as a formal mathematics project, you would title it something like **"The Immersion Dimension Project"**.

- **Immediate action:** You can immediately start writing the Lean 4 definitions for $\alpha(n)$ and formally proving combinatorial properties about bounds for binary expansions.
- **Mathlib contribution:** You could work on the formal definition of Stiefel-Whitney classes via the Steenrod algebra, which is currently an open, highly desired target in the Lean topology community.
- **The ultimate synthesis:** Once characteristic classes are in Lean, you can formally prove that $\mathbb{RP}^2$ (the projective plane) cannot immerse in $\mathbb{R}^2$, and $\mathbb{RP}^4$ cannot immerse in $\mathbb{R}^6$, successfully bridging the algebraic geometry of Text 1 with the deep differential topology of Text 2 inside a machine-verified environment.